electricity, fuel, water, land, precious minerals and other resources are used to manufacture and operate them. Data centres alone account for between 1 per cent and 1.5 per cent of global energy usage, and large data centres take up millions of square feet and require hundreds of thousands of gallons of fresh water every day to keep them from overheating.[46]

I also use the term ‘algorithm’, when I wish to focus more on software aspects, but it is crucial to remember that all the algorithms mentioned in subsequent pages run on some computer or other. As for the term ‘AI’, I use it when emphasising the ability of some algorithms to learn and change by themselves. Traditionally, AI has been an abbreviation for ‘artificial intelligence’. But for reasons already evident from the previous discussion, it is perhaps better to think of it as ‘alien intelligence’. As AI evolves, it becomes less artificial (in the sense of depending on human designs) and more alien. It should also be noted that people often define and evaluate AI through the metric of ‘human-level intelligence’, and there is much debate about when we can expect AIs to reach ‘human-level intelligence’. The use of this metric, however, is deeply confusing. It is like defining and evaluating aeroplanes through the metric of ‘bird-level flight’. AI isn’t progressing toward human-level intelligence. It is evolving an entirely different type of intelligence.

Another confusing term is ‘robot’. In this book it is used to allude to cases when a computer moves and operates in the physical sphere, whereas the term ‘bot’ refers to algorithms operating mainly in the digital sphere. A bot may be polluting your social media account with fake news, while a robot may clean your living room of dust.

One last note on terminology: I tend to speak of the computer-based ‘network’ in the singular, rather than about ‘networks’ in the plural. I am fully aware that computers can be used to create many networks with diverse characteristics, and chapter 11 explores the possibility that the world will be divided into radically different and even hostile computer networks. Nevertheless, just as different tribes, kingdoms and churches share important features that enable us to talk about a single human network that has come to dominate planet Earth, so I prefer to talk about *the* computer network in the singular, in order to contrast it to the human network it is superseding.